ApproxDA-TransUNet: Understanding Context Sensitivity of Attention Approximation in Medical Image Segmentation

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***Abstract*—We study Context Sensitivity (CS), a task-level property measuring how much a task's segmentation performance changes with the spatial receptive field of attention, operationalized as the DSC range across a window size ablation. We hypothesize that CS governs the importance of window selection: high-CS tasks require careful window tuning to achieve gains; low-CS tasks are window-robust. To validate this, we design ApproxDA-TransUNet, a controllable dual-attention architecture with three orthogonal approximation operators—windowed PAM (spatial scope M), low-rank projection (representation fidelity r), and grouped CAM (channel scope G)—reducing per-GPU training memory from 11.5 GB to 6.4 GB and enabling multi-GPU distributed training. Experiments on three benchmarks confirm the CS hypothesis: with task-appropriate window selection, ApproxDA improves both multi-organ CT segmentation (+1.14% vs. DA-TransUNet on Synapse with M=28; 80.94% DSC, CS=2.30) and polyp segmentation (+1.73% on Kvasir-SEG with M=56; 90.17% DSC, CS=0.64)—a 3.6× sensitivity ratio confirming CS as a predictor of window-tuning importance. The DSC-vs-M curve on Synapse is non-monotonic (peak at M=28), directly paralleling the bias-variance tradeoff: under-context windows have high bias; over-context windows introduce spurious variance; the intermediate optimum balances both. A secondary finding reveals gradient symmetry collapse: learnable dual-branch gating converges to a near-constant $g{\approx}0.5$ regardless of window size, confirmed at both M=7 and M=14. These results shift the research question from *how to approximate* to *when, and at what scale*, with CS providing a practical task-level criterion.**

***Index Terms*—medical image segmentation, dual attention, efficient transformers, attention approximation, context sensitivity, gate collapse**

I. Introduction

Transformers have revolutionized medical image segmentation by enabling global long-range dependency modeling that convolutional networks cannot achieve [11], [12]. Dual attention [1]—jointly modeling positional and channel correlations—emerged as a powerful mechanism, and methods such as DA-TransUNet [2] demonstrated its effectiveness in hybrid CNN-Transformer architectures for multi-organ and polyp segmentation. The quadratic complexity of dual attention, however, has prompted a wave of efficient approximations—windowed attention [3], low-rank factorization [4], and grouped channel interaction [5]—that are now standard design choices. These strategies share an implicit assumption that approximation is universally applicable with minor accuracy trade-offs—an assumption rarely examined across tasks with fundamentally different spatial reasoning requirements:

> *When, and at what scale, is attention approximation beneficial for medical image segmentation?*

To investigate this, we design **ApproxDA-TransUNet**: a dual-attention architecture with three orthogonal, independently controllable approximation operators—windowed PAM, low-rank projection, and grouped CAM—serving as a scientific instrument that isolates each operator's contribution across tasks.

Experiments reveal a consistent task-dependent pattern: appropriate window selection yields gains on both multi-organ CT and polyp segmentation, but sensitivity to window choice differs markedly across tasks. We further find that learnable routing collapses to a fixed equilibrium due to gradient symmetry—a structural property of symmetric two-branch architectures, not a training artifact.

These findings motivate **Context Sensitivity (CS)**, a task-level concept operationalized as the performance range across window sizes. High-CS tasks require careful window tuning to realize gains; low-CS tasks exhibit broad performance plateaus and are window-robust. CS explains the observed task-dependent behavior and guides attention strategy selection.

The main contributions of this paper are:

- **ApproxDA-TransUNet**: a controllable three-axis approximation framework enabling isolation of windowed PAM, low-rank projection, and grouped CAM effects across segmentation tasks with different contextual requirements.
- **Context Sensitivity (CS)**: a task-level concept governing window selection importance, validated by a non-monotonic DSC-vs-window curve on high-CS tasks that mirrors the bias-variance tradeoff.
- **Gradient symmetry collapse**: an analysis showing that symmetric dual-branch gating converges to $g{\approx}0.5$ regardless of window size or rank—a fundamental limitation of

symmetric two-branch attention design.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the ApproxDA framework; Section IV reports experimental results; Section V provides discussion and analysis; Section VI concludes.

II. Related Work

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures established the foundational paradigm for medical image segmentation, with U-Net [6] and its successors [7]–[10] progressively refining skip connections, attention gating, and multi-scale feature representations. Despite strong performance on local-feature-dominant tasks, their limited receptive fields hinder long-range anatomical reasoning.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [11], Swin-Unet [12], UNETR [13], UCTransNet [14], and DAEformer [15] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks.

C. Dual Attention and DA-TransUNet

DANet [1] introduced Position Attention Modules (PAM) and Channel Attention Modules (CAM), capturing spatial and channel correlations at $\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$ complexity respectively. DA-TransUNet [2] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections of a hybrid R50-ViT-B/16 encoder-decoder, achieving state-of-the-art accuracy while inheriting the quadratic cost of full-map attention.

D. Efficient Attention and Adaptive Routing

Windowed local attention [3], low-rank projection [4], and grouped channel interaction [5] have become standard strategies for reducing the cost of global attention, with learnable gating widely adopted to fuse multi-branch outputs [5], [15]. Prior work focuses on *how* to approximate efficiently, leaving open *when* approximation is appropriate; this work addresses both by studying task-dependent approximation behavior and exposing the gradient symmetry dynamics that cause adaptive routing to degenerate into fixed blending.

III. Method

This section introduces the overall architecture of ApproxDA, then describes the three controllable approximation operators used to reformulate the dual-attention module, and finally analyzes their computational complexity.

Fig. 1. ApproxDA-TransUNet architecture. ApproxDABlocks (Low-Rank Windowed PAM + Grouped CAM + gate routing) replace dual-attention modules at all skip connections and the bottleneck. `gate_mode` controls the routing during ablation.

A. Architecture Overview

ApproxDA-TransUNet adopts the hybrid CNN–ViT encoder-decoder structure of DA-TransUNet [2], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at 28×28, 256 at 56×56, and 64 at 112×112—and at the bottleneck.

ApproxDA-TransUNet is designed around an explicit hypothesis: that three orthogonal approximation operators—windowed PAM, low-rank projection, and grouped CAM—can replace the quadratic DA module to yield both a practical efficient architecture and a systematic framework for studying when approximation is appropriate. Each operator targets a distinct approximation axis (spatial scope, representation fidelity, channel scope), so any two can be held fixed while the third varies. This makes ApproxDA not merely a faster model, but a model whose design enables controlled ablation of approximation effects across tasks with different contextual requirements.

Figure 1 illustrates the overall ApproxDA architecture.

B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity in both spatial resolution and channel dimension. ApproxDA reduces this cost along three independent axes, each targeting a distinct aspect of the approximation:

Spatial scope (window size M) controls *how far* PAM reaches. Full global attention ($M{=}H$) captures all long-range spatial relationships; restricting to $M\times M$ local windows trades long-range context for lower complexity. This axis most directly governs whether high-GCR tasks retain the anatomical context they require.

Representation fidelity (rank r) controls *how precisely* attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher r approaches exact attention.

Semantic interaction (groups G) controls *which channels* may interact in CAM. Partitioning C channels into G independent groups reduces cross-channel dependency while retaining within-group semantic interaction.

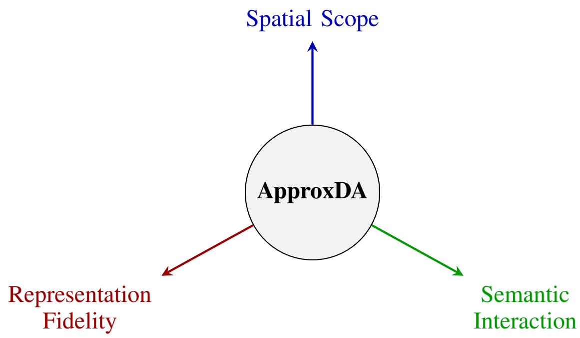

Fig. 2. ApproxDA's three-dimensional approximation space. Each axis is independently controllable. Spatial scope (M) most directly governs long-range context capture and thereby approximation effectiveness on high-GCR tasks.

These axes correspond to three distinct types of approximation: *spatial approximation* (how far attention reaches, controlled by M), *representation approximation* (how precisely affinity is computed, controlled by r), and *channel approximation* (how broadly channels interact, controlled by G). The axes are fully independent: any two of $\{M, r, G\}$ can be held fixed while the third varies, enabling clean ablations that isolate individual approximation effects. Figure 2 visualizes this three-dimensional approximation space.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA partitions the feature map into non-overlapping windows of size $M\times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

Window partitioning. Following [3], we partition the feature map into n_W non-overlapping $M\times M$ windows. Window clamping $M \leftarrow \min(M, H, W)$ allows M=112 to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

Low-rank projection. Following [4], query and key features inside each window are projected from N_w=M^2 to rank $r \ll M^2$ via a learned matrix $\mathbf{W}_r \in \mathbb{R}^{r\times N_w}$:

$$\tilde{\mathbf{C}} = \mathbf{W}_r\mathbf{C}^\top, \quad \tilde{\mathbf{D}} = \mathbf{W}_r\mathbf{D}^\top \in \mathbb{R}^{r\times C}. \tag{1}$$

The attention matrix $QK^\top \in \mathbb{R}^{N_w\times N_w}$ is approximated by $Q_rK_r^\top \in \mathbb{R}^{N_w\times r}$, where $Q_r, K_r \in \mathbb{R}^{N_w\times r}$. Total complexity is $\mathcal{O}(NrC)$. Combined, the two approximations reduce attention storage at 112×112 from ≈15 GB to ≈38 MB (∼390×).

These two hyperparameters—window size M and projection rank r—jointly constitute the spatial approximation operator and form the primary experimental axes investigated on Synapse.

D. Channel Approximation

The original Channel Attention Module computes a $C\times C$ channel attention matrix at $\mathcal{O}(C^2N)$ cost. ApproxDA approximates this by partitioning C channels into G groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)}\cdot\mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g}\exp(\mathbf{A}_k^{(g)}\cdot\mathbf{A}_j^{(g)})}, \tag{2}$$

with total cost $\mathcal{O}(C^2N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependency.

The grouping number G provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

E. Routing Strategy

The learnable routing mechanism of DA-TransUNet [2] is retained and extended to a configurable `gate_mode`. The outputs of the spatial and channel attention branches are fused via:

$$\mathbf{y} = \mathbf{W}_f\left(g\cdot\hat{\mathbf{P}} + (1-g)\cdot\hat{\mathbf{C}}\right) + \mathbf{x}, \tag{3}$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a 1×1 convolution, and g is determined by the `gate_mode` parameter:

`pam`:	$g = 1.0$	(PAM only)
`cam`:	$g = 0.0$	(CAM only)
`fixed`:	$g = 0.5$	(equal blend)
`learn`:	$g = \sigma(\mathbf{W}_g\cdot\mathrm{GAP}(\mathbf{x})) \in (0,1)^C$	

This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA does not treat this routing module as a primary architectural contribution; instead, it serves as an analyzable component whose optimization behavior is investigated in Section V. As demonstrated later, the routing coefficient consistently converges to an almost constant value, revealing an unexpected optimization phenomenon.

F. Loss Function

The training objective combines Dice loss and cross-entropy loss with equal weights:

$$\mathcal{L} = \tfrac{1}{2}\mathcal{L}_{\text{Dice}} + \tfrac{1}{2}\mathcal{L}_{\text{CE}}. \tag{4}$$

These approximations reduce layer-level attention complexity: windowed PAM from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NrC)$ ($r \ll N$); grouped CAM from $\mathcal{O}(C^2N)$ to $\mathcal{O}(C^2N/G)$. Since backbone operations dominate end-to-end computation, the primary value of ApproxDA is not whole-network FLOPs reduction but a controllable formulation for systematically isolating each approximation operator's effect on segmentation performance (Section IV).

IV. Experiments

We evaluate ApproxDA-TransUNet on three medical image segmentation benchmarks spanning different context sensitivity. The experiments systematically examine computational efficiency, task-dependent approximation behavior across CS regimes, the dynamics of symmetric gate routing, and the relationship between approximation strength and task context.

TABLE I
DATASET PROPERTIES AND GLOBAL CONTEXT REQUIREMENT (CS)

Dataset	Classes	Object Type	Long-range Context	CS
Synapse	8	Multi-organ CT	High (inter-organ)	High
Kvasir-SEG	1	Polyp	Low (local boundary)	Low
ISIC 2018	1	Skin lesion	Low (texture/edge)	Low

High-CS tasks require cross-organ anatomical reasoning; low-CS tasks are dominated by local appearance cues. CS governs approximation effectiveness (Section V).

A. Experimental Setup

Datasets. We evaluate ApproxDA on three representative medical image segmentation benchmarks covering different contextual characteristics.

Synapse Multi-Organ CT [16]: 30 abdominal CT scans with an 18/12 train/test split covering 8 organ classes (9 classes with background). Evaluation metrics: mean DSC and HD95.

Kvasir-SEG [17]: 1000 endoscopy polyp images (binary segmentation) with an 800/200 train/test split. Evaluation metrics: DSC and mIoU.

ISIC 2018 [18]: 2594 dermoscopy skin lesion images (binary segmentation) with a 2075/519 train/test split (80/20). Evaluation metrics: DSC and mIoU.

Table I summarizes the three datasets alongside their contextual characteristics and CS classification, which motivates the cross-task analysis in Section V.

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Training uses SGD (momentum 0.9, weight decay 10^{-4}), initial learning rate 0.01 with polynomial decay, batch size 24, image size 224×224, 300 epochs, and seed 1234. The DA-TransUNet baseline runs on T4×1. ApproxDA-TransUNet runs on T4×2 via `torchrun` DDP with per-GPU batch 12 and NCCL all-reduce, under identical total batch size and base learning rate. Default ApproxDA configuration: M=7, r=32, G=8. Our DA-TransUNet re-run achieves validation DSC 0.795, consistent with the paper-reported 0.798, confirming faithful replication.

B. Computational Efficiency of ApproxDA

Table II compares the computational characteristics of ApproxDA and DA-TransUNet on Synapse. Although per-GPU VRAM is reduced substantially—from 11.5 GB to 6.4 GB—total GFLOPs are slightly higher (32.1 vs. 30.2) because the shared ViT backbone dominates total computation and Conv1d/Linear projection overhead partially offsets decoder-level savings. The reduced memory footprint enables multi-GPU distributed training via DDP, which DA-TransUNet does not support.

C. Task-dependent Behavior of Attention Approximation

We examine whether efficient attention approximation behaves consistently across medical segmentation tasks with different contextual requirements. Tables III–V compare ApproxDA-TransUNet against DA-TransUNet and established baselines on all three benchmarks.

Synapse. Table III compares DSC and HD95 against 11 published methods. CNN-based methods achieve 76–78% DSC. Transformer-based methods progressively improve global context modeling, with Swin-Unet (79.13%) and DA-TransUNet (79.80%) previously topping the leaderboard. ApproxDA-TransUNet (gate=pam, M=28, r=32) achieves **80.94%** DSC and 27.49 mm HD95—surpassing all baselines including DA-TransUNet by $+1.14\%$. Window sensitivity ablation (Section IV-D) reveals a non-monotonic DSC-vs-M curve peaking at M=28: both too-local (M=7: 78.64%) and fully-global (M=112: 79.44%) attention underperform the intermediate optimum.

Kvasir-SEG. Table IV compares polyp segmentation results against all six methods reported in [2]. We re-ran DA-TransUNet under the same SGD/300ep protocol as ApproxDA to enable a direct comparison, obtaining 88.44% DSC, 81.70% mIoU, and 53.04 mm HD95. ApproxDA-TransUNet (gate=pam, M=56) achieves 90.17% DSC, 84.27% mIoU, and 44.35 mm HD95—outperforming the reproduced DA-TransUNet baseline (+1.73% DSC, +2.57% mIoU, -8.69 mm HD95).

ISIC 2018. Table V reports skin lesion segmentation results against the same six baselines from [2]. Our ApproxDA experimental results are pending completion.

With appropriate window selection, ApproxDA outperforms DA-TransUNet on both datasets: $+1.14\%$ DSC on Synapse (M=28) and $+1.73\%$ DSC on Kvasir (M=56). However, the sensitivity to window choice differs markedly: Synapse DSC spans 2.30% across M values while Kvasir spans only 0.64%, a 3.6× ratio reflecting the different context sensitivity of the two tasks.

D. Window Sensitivity and the Optimal Approximation Scale

Which approximation axis drives performance, and what is the optimal scale? Table VI isolates spatial windowing by comparing $M\in\{7, 28, 112\}$ under gate=pam, r=32 on Synapse. Crucially, *approximation is present in all rows*—windowed PAM and low-rank projection operate at every M, including M=28—demonstrating that optimal performance requires the right approximation scale, not the absence of approximation.

The Synapse DSC-vs-M curve is *non-monotonic*: peak at M=28 (80.94%, $+1.14\%$ vs. DA-TransUNet), with both too-local (M=7: 78.64%) and fully-global (M=112: 79.44%) attention underperforming. This mirrors the classical bias-variance tradeoff: under-context windows have high bias (missing necessary dependencies); over-context windows introduce spurious variance (irrelevant distant tokens); the intermediate M=28 achieves the optimal balance. The DSC range of 2.30% across M values confirms high window sensitivity on this high-CS task.

The Kvasir-SEG curve ($M\in\{7, 28, 56, 112\}$) spans only 0.64% DSC—a broad performance plateau with a shallow peak at M=56. Even on this low-CS task, moderate global

TABLE II
EFFICIENCY COMPARISON ON SYNAPSE (T4). GFLOPS MEASURED BY `FVCORE` (SINGLE 224×224 SLICE, INCLUDES ATTENTION `BMM`).

Method	Params (M)	GFLOPs	Train VRAM	Train Time	Infer (s/vol)	Multi-GPU
DA-TransUNet [2]	107.95	30.2	11.5 GB	11.4h ×1	120.0	No†
ApproxDA (gate=pam, M=7, r=32)	112.98	32.1	6.4 GB/GPU	8.3h ×2	121.3	Yes (DDP)
ApproxDA (gate=learn, M=7, r=32)	114.90	32.1	10.6 GB	11.5h ×1	114.2	Yes (DDP)

†DA-TransUNet does not support multi-GPU training. ApproxDA uses DDP (`torchrun`, NCCL); per-GPU batch 12 (total 24), LR unchanged.

TABLE III
SEGMENTATION RESULTS ON SYNAPSE MULTI-ORGAN CT

Method	DSC (%) ↑	HD95 (mm) ↓
CNN-based		
U-Net [6]	76.85	39.70
U-Net++ [7]	76.91	36.93
Residual U-Net [9]	76.95	38.44
Att-Unet [8]	77.77	36.02
MultiResUNet [10]	77.42	36.84
Transformer-based		
TransUNet [11]	77.48	31.69
UCTransNet [14]	78.23	26.75
TransNorm [19]	78.40	30.25
MIM [20]	78.59	26.59
Swin-Unet [12]	79.13	**21.55**
DA-TransUNet [2]	<u>79.80</u>	<u>23.48</u>
Dual-attention approximation (ours)		
ApproxDA-TransUNet	**80.94**	27.49

Baselines from [2] Table 1 (SGD, 500ep, 224×224). ApproxDA: gate=pam, M=28, r=32, G=8, SGD 300ep (window sensitivity ablation identifies M=28 as optimal). We also re-ran DA-TransUNet under our protocol (SGD, 300ep) and obtained 72.03% DSC; the gap from paper-reported (79.80%) is primarily due to optimizer and epoch differences (Adam/500ep vs. SGD/300ep), so we report paper-reported values for all Synapse baselines.

TABLE IV
SEGMENTATION RESULTS ON KVASIR-SEG POLYP

Method	DSC (%) ↑	mIoU (%) ↑	HD95 (mm) ↓
U-Net [6]	88.22	80.12	–
Att-Unet [8]	86.61	78.01	–
U-Net++ [7]	86.57	77.67	–
Res-UNet [9]	77.85	66.04	–
TransUNet [11]	87.91	80.03	–
DA-TransUNet [2]†	<u>88.44</u>	<u>81.70</u>	<u>53.04</u>
ApproxDA-TransUNet (ours)	**90.17**	**84.27**	**44.35**

Other baselines from [2] Table 2 (Adam, 500ep, 75/25 split); HD95 not reported. †DA-TransUNet re-run under the same protocol as ApproxDA (SGD, 300ep, 80/20 split). Paper-reported [2]: 88.47% DSC, 81.02% mIoU. ApproxDA: gate=pam, M=56, r=32.

TABLE V
SEGMENTATION RESULTS ON ISIC 2018 SKIN LESION

Method	DSC (%) ↑	mIoU (%) ↑
U-Net [6]	87.22	81.14
Att-Unet [8]	87.60	81.51
U-Net++ [7]	87.30	81.33
Res-UNet [9]	83.32	76.51
TransUNet [11]	<u>88.78</u>	<u>82.63</u>
DA-TransUNet [2]	**88.88**	**82.78**
ApproxDA-TransUNet (ours)	*pend.*	*pend.*

Baselines from [2] Table 2 (Adam, 50ep, 75/25 split). ApproxDA: SGD, 300ep, 80/20 split.

TABLE VI
SPATIAL APPROXIMATION ABLATION ON SYNAPSE (G=8)

Config	M	r	Gate	DSC (%) ↑	HD95 (mm) ↓
Windowed + low-rank	7	32	pam	78.64	31.09
Windowed + low-rank	14	64	learn†	78.04	29.09
Windowed + low-rank	28	32	pam	**80.94**	<u>27.49</u>
Global + low-rank	112	32	pam	<u>79.44</u>	**27.26**

†Gate collapsed to $g{\approx}0.5$. $M{=}112$ via window clamping yields full-map attention at all scales. Non-monotonic peak at $M{=}28$: both under-context ($M{=}7$) and over-context ($M{=}112$) underperform. DSC range = 2.30%—3.6× Kvasir's 0.64%, operationalizing CS as a sensitivity predictor.

context ($M{=}56$) edges out fully-local ($M{=}7$) attention, but the task is largely window-robust across the measured range.

Figure 3 visualizes both curves side by side. The 3.6× sensitivity ratio directly reflects the tasks' different context sensitivity.

Cross-task summary. Table VII consolidates results. With task-appropriate window selection, ApproxDA improves over DA-TransUNet on both benchmarks: +1.14% on high-CS Synapse and +1.73% on low-CS Kvasir. The sensitivity ratio (2.30% vs. 0.64%) is the primary empirical operationalization of CS. Figure 4 shows qualitative examples.

Together, the window sensitivity experiments establish that approximation strength—not approximation itself—governs performance. This motivates the mechanistic analysis in Section V.

E. Analysis of Symmetric Gate Collapse

Although the learnable gate (`gate=learn`) was designed to dynamically balance PAM and CAM branches, Table VIII reveals that it consistently ranks last on Synapse. The gate coefficient converges to $g{\approx}0.5$ and remains nearly constant throughout training, producing a fixed 50/50 blend that performs below both PAM-only (78.64%) and CAM-only (78.26%) configurations. This collapse is observed across both $M{=}7$ and $M{=}14$, confirming that it is a property of symmetric two-branch optimization rather than a consequence

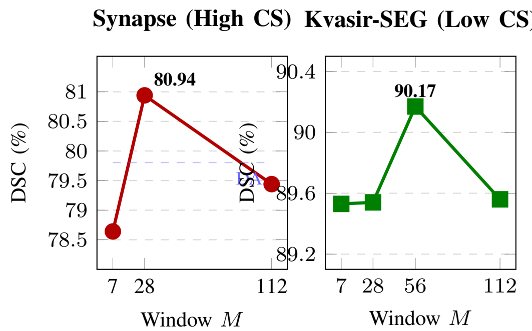

Fig. 3. DSC vs. window size M (gate=pam, r=32). **Left (Synapse, high CS):** non-monotonic curve, peak at M=28 (+1.14% vs. DA-TransUNet, dashed blue), range 2.30%. **Right (Kvasir-SEG, low CS):** near-flat plateau, range 0.64%. The 3.6× sensitivity ratio operationalizes CS empirically.

Fig. 4. Qualitative segmentation comparison. **Top (Synapse, M=28):** ApproxDA achieves +1.14% DSC; performance is window-sensitive (range 2.30%). **Bottom (Kvasir-SEG, M=56):** ApproxDA achieves +1.73% DSC; performance is window-robust (range 0.64%). [Generated after experiments complete.]

TABLE VII
CROSS-TASK APPROXIMATION EFFECTIVENESS BY CS LEVEL

Dataset	CS	DA-TransUNet[†]	Best ApproxDA	Δ DSC
Synapse	High	79.80%	80.94%[a]	+1.14%
Kvasir-SEG	Low	88.44%[c]	90.17%[b]	+1.73%
ISIC 2018	Low	88.88%	*pend.*	*pend.*

[†]Paper-reported [2] except [c]Kvasir (our re-run, SGD 300ep 80/20). [a]gate=pam, M=28, r=32. [b]gate=pam, M=56, r=32.

TABLE VIII
GATE MODE ABLATION ON SYNAPSE (M=7, r=32 UNLESS NOTED)

Gate Mode	g	DSC (%) ↑	HD95 (mm) ↓
gate=pam	1.0	**78.64**	31.09
gate=cam	0.0	<u>78.26</u>	<u>30.59</u>
gate=learn (M=14, r=64)	≈0.5*	78.04	**29.09**
gate=learn	≈0.5*	77.78	34.29

All rows: ApproxDA-TransUNet. *Gate collapsed to g≈0.5 throughout training; see Section V-A. **Bold**: best; <u>underline</u>: second best per column.

of limited receptive field. The gradient-level derivation is provided in Section V-A.

V. DISCUSSION

A. Observation 1: Gate Collapse

Derivation. Given fused output $\mathbf{F} = g\mathbf{F}_{\mathrm{P}} + (1-g)\mathbf{F}_{\mathrm{C}}$, the gate gradient is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_{\mathrm{P}} - \mathbf{F}_{\mathrm{C}})\,. \tag{5}$$

At initialization g≈0.5, both branches receive equal gradient $\frac{1}{2}\partial\mathcal{L}/\partial\mathbf{F}$. Without incentive to differentiate, $\mathbf{F}_{\mathrm{P}} \approx \mathbf{F}_{\mathrm{C}}$, so Eq. (5)≈ $\mathbf{0}$. This is self-reinforcing: equal gradients → convergent representations → near-zero gate gradient → g fixed at 0.5. This represents a **stable equilibrium of symmetric dual-branch optimization**: once reached, no first-order gradient signal can escape it.

Empirical confirmation. After 300 epochs, the gate range at M=7 is [0.4993, 0.5010] with Δg=0.0017. At M=14: g≈0.5002 with Δg≈0.0000. Collapse is identical at both window sizes, confirming it is a fundamental property of symmetric two-branch gating and not a consequence of limited receptive field. Figure 5 visualizes the collapse trajectory.

Design implication. Symmetric two-branch gating is an unstable fixed point that any dual-attention architecture will converge to without explicit symmetry breaking. Future designs should initialize branches asymmetrically, use diversity-encouraging loss terms, or replace the gate with a non-symmetric routing mechanism to escape this degenerate equilibrium.

B. Observation 2: Task-Dependent Approximation Effectiveness

Although ApproxDA adopts the same approximation operators across all experiments, its effectiveness varies considerably among different segmentation datasets. Window sensitivity ablation (Phase D, gate=pam, r=32, $M \in \{7, 28, 56, 112\}$) reveals that performance depends critically on window selection, and that the optimal window size differs by task: M=28 on Synapse (80.94% DSC, +1.14% vs. DA-TransUNet) and M=56 on Kvasir-SEG (90.17% DSC, +1.73%). Importantly, the *sensitivity* to window choice differs dramatically: the Synapse DSC-vs-M curve spans 2.30% across measured M values (non-monotonic: M=7 78.64%, M=28 80.94%, M=112 79.44%), while the Kvasir curve spans only 0.64%—a 3.6× ratio.

We attribute this sensitivity difference to the different global contextual requirements of the underlying segmentation tasks. Multi-organ abdominal CT segmentation requires reasoning over long-range anatomical relationships among multiple organs. The high sensitivity to window size (2.30% range) reflects that both too-local windows (insufficient cross-organ context) and fully-global windows (noise from unrelated background) degrade performance; an intermediate scope (M=28) provides the optimal balance.

In contrast, polyp and skin lesion segmentation are primarily determined by local appearance, boundary contrast, and texture consistency. The performance improvement on Kvasir-SEG should not be interpreted as approximation creating additional

Fig. 5. Gate value g vs. training epoch for $M{=}7$ and $M{=}14$. Both converge to $g{\approx}0.5$ by epoch 100, independent of window size. Shaded band shows gate range across network blocks. [Generated from training logs after experiments complete.]

Fig. 6. Gate value vs. per-block input entropy scatter (300-epoch checkpoint, Synapse validation). Gate values cluster tightly at $g{\approx}0.5$ regardless of input entropy, confirming no sample-adaptive routing occurs. [Generated from inference analysis.]

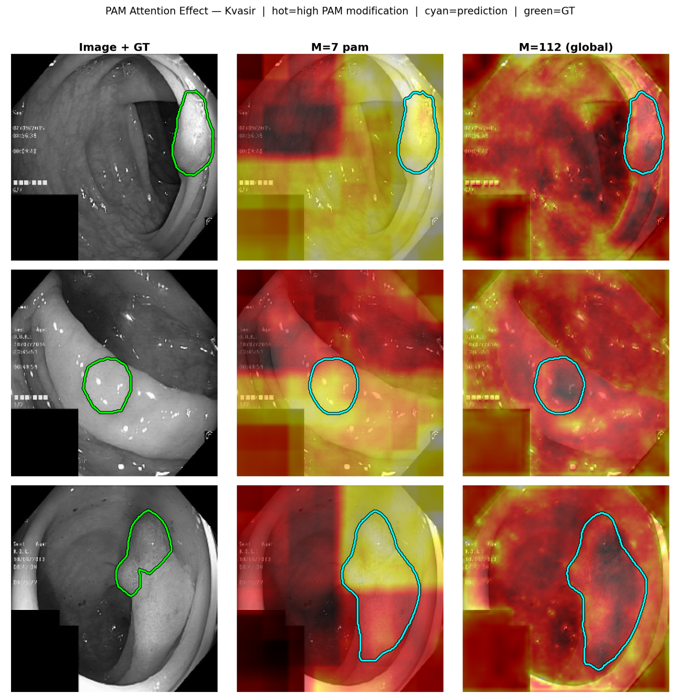

Fig. 7. Attention-map analysis on three representative Kvasir-SEG polyp images (selected from 10; samples with largest M=7 vs. M=112 on-mask gap). Each row: input image with GT contour (left), windowed PAM heatmap ($M{=}7$, gate=pam; center), global PAM heatmap ($M{=}112$, gate=pam; right). Windowed attention concentrates 62.3% of its effect on the polyp region vs. 34.2% for global attention—a $1.82\times$ difference confirmed in $9/10$ samples. Heatmap intensities reflect $\|\text{output} - \text{input}\|_2$ averaged across decoder scales and upsampled to $224{\times}224$.

discriminative information. Rather, we hypothesize that attention approximation introduces an *inductive bias* favoring local interactions while suppressing unnecessary long-range dependencies. For low-CS tasks, where segmentation is primarily governed by local boundary appearance and texture, this bias better matches the intrinsic structure of the problem and may simultaneously act as an implicit regularizer. The near-flat curve (0.64% range) confirms low context sensitivity: any reasonable window achieves competitive performance. We validate the locality bias hypothesis via controlled attention-map analysis: comparing gate=pam $M{=}7$ (windowed) versus $M{=}112$ (global) attention effects on 10 Kvasir polyp images, windowed attention concentrates 62.3% of its effect on the polyp region versus only 34.2% for global attention—a $1.82\times$ difference confirmed in $9/10$ samples (Figure 7). Windowed attention focuses on polyp boundaries while global attention disperses to surrounding tissue, providing direct evidence for the inductive bias alignment hypothesis. This interpretation motivates the more precise characterization below.

C. Why Does Approximation Improve Performance?

Restricting the attention receptive field does not merely reduce computational cost—it changes the model's *inductive bias*. Three complementary mechanisms explain why windowed PAM outperforms global attention at the appropriate scale.

Inductive bias alignment. Global PAM allows every token to attend to every other, providing theoretical maximum flexibility. For tasks where relevant information is predominantly local—polyp boundaries, skin lesion edges—this unconstrained connectivity causes attention to diffuse toward irrelevant background. Windowed PAM enforces a locality prior that matches the task's intrinsic structure. The attention-map analysis (Section V-B) provides direct evidence: $M{=}7$ PAM concentrates 62.3% of its effect on the polyp region vs. only 34.2% for $M{=}112$ global attention—a $1.82\times$ improvement in on-target signal. This parallels the success of CNNs: local connectivity is not a limitation but a task-aligned prior.

Over-context suppression. The Synapse DSC-vs-M curve is non-monotonic even on a high-CS task: performance *decreases* from $M{=}28$ to $M{=}112$ (80.94% $\rightarrow$ 79.44%). Fully-global attention introduces spurious long-range correlations from anatomically unrelated regions, acting as noise. An intermediate window captures sufficient cross-organ context while suppressing background interference. This is a direct analogue of the bias-variance tradeoff: under-context (small M) yields high bias (missing dependencies); over-context (large M) yields high variance (spurious correlations); the optimal intermediate M minimizes both.

Implicit regularization. Constraining the receptive field reduces the effective degrees of freedom in the attention mechanism, which may act as implicit regularization—suppressing overfitting to spurious patterns in limited medical training sets.

This mechanism parallels dropout or weight decay; validating it directly requires comparing train-DSC vs. test-DSC learning curves, which we leave for future work.

Together, these mechanisms reframe the role of approximation: not as a *reduction of capability* but as an *inductive bias adjustment*. The optimal window is the one that best aligns the attention scope with the task's intrinsic context scale.

D. Context Sensitivity (CS) and the Approximation Spectrum

Context Sensitivity (CS) is the degree to which a task's segmentation performance changes with the effective spatial context available to the attention mechanism. We operationalize CS empirically as the DSC range across the window sensitivity ablation:

$$\mathrm{CS} \triangleq \max_{M} \mathrm{DSC}(M) - \min_{M} \mathrm{DSC}(M), \tag{6}$$

measured under fixed gate=pam, r=32. Under this definition, CS(Synapse) $= 80.94 - 78.64 = 2.30$ percentage points; CS(Kvasir) $= 90.17 - 89.53 = 0.64$ percentage points—a 3.6× ratio. High-CS tasks exhibit steep DSC-vs-M curves with a well-defined optimal scale; low-CS tasks show a broad performance plateau where the choice of M matters little. Crucially, high CS does not imply approximation is harmful: with the right window, ApproxDA outperforms full dual attention on *both* tasks. CS determines how *sensitive* performance is to window choice—not the direction of the effect.

We propose that CS governs the *importance* of window tuning, which we treat as an empirical hypothesis rather than a proven causal law:

$$\Delta_{\mathrm{DSC}} \approx f(\mathrm{CS}), \quad f'(\cdot) < 0, \tag{7}$$

where Δ_{DSC} = DSC(ApproxDA) − DSC(DA-TransUNet) at the *optimal window* for each task. Equation (7) is a conceptual claim, not an analytic proof. With task-appropriate windows, ApproxDA achieves Δ_{DSC}=+1.73% on low-CS Kvasir-SEG (M=56) and Δ_{DSC}=+1.14% on high-CS Synapse (M=28)—both positive, with larger gain on the window-robust low-CS task. Figure 8 illustrates the CS-governed pattern. The 3.6× sensitivity ratio operationalizes the CS difference: high-CS tasks require careful window tuning to unlock gains; low-CS tasks are window-robust. Systematic multi-task CS quantification is a direction for future work.

Attention approximation should not be regarded as a binary design choice but as a configurable inductive bias whose importance depends on task CS. Different window sizes suit different CS regimes (Figure 2). That the same collapsed gate=learn routing helps on low-CS Kvasir (+0.77% vs. DA-TransUNet at M=7) while underperforming gate=pam on high-CS Synapse (+1.14% with tuned M=28) further underscores this task-dependence: the effectiveness of a fixed routing strategy is determined by the task's context sensitivity.

E. Design Principles

The CS framework and the gate collapse finding together provide three practical principles for efficient attention design in medical image segmentation.

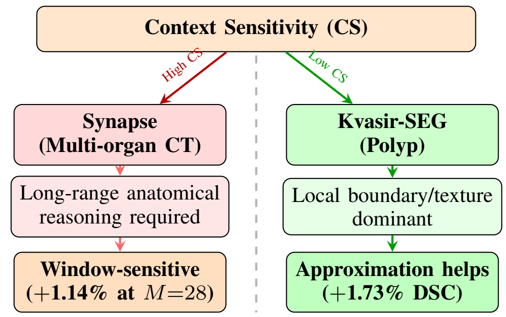

Fig. 8. Context Sensitivity (CS) governs approximation window importance. Both tasks benefit from ApproxDA with the appropriate window (+1.14% Synapse at M=28; +1.73% Kvasir at M=56), but performance sensitivity to window choice is 3.6× higher on the high-CS benchmark (Synapse CS=2.30 pp vs. Kvasir CS=0.64 pp).

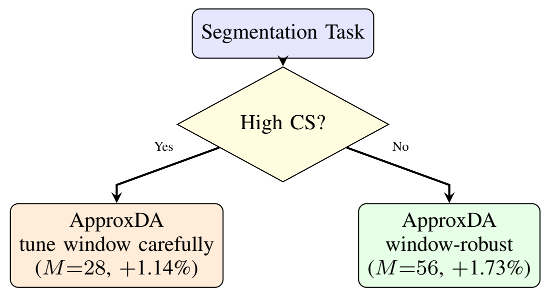

Fig. 9. Practical design guideline derived from CS analysis. ApproxDA outperforms DA-TransUNet on both task types with appropriate window selection. High-CS tasks require careful window tuning (M=28, +1.14%); low-CS tasks are window-robust (M=56, +1.73%). Both cases reduce per-GPU training VRAM by 44% (6.4 GB vs. 11.5 GB).

First, attention approximation should be selected according to the contextual characteristics of the target segmentation task, rather than applied uniformly across all medical imaging applications.

Second, adaptive routing mechanisms require sufficient feature diversity between attention branches; otherwise, gradient symmetry may cause routing collapse regardless of window size or rank.

Finally, future efficient attention architectures should explicitly consider when approximation is appropriate—not just how to make it efficient—rather than focusing exclusively on reducing theoretical complexity. Figure 9 summarizes these guidelines as a practical decision tree.

F. Limitations

Although Eq. (6) provides a computable CS value from the window sensitivity ablation, a rigorous CS estimator that generalizes across imaging modalities and training protocols—without requiring the ablation itself—remains to be developed. A natural candidate proxy, center-cropping the input to restrict visible context, is invalidated by a more fundamental issue: evaluation against the full-resolution label penalises the crop for missing structures that lie outside the crop window, regardless of the model's actual context reasoning. For randomly-positioned targets (polyps, lesions), small crops frequently exclude the target entirely; for anatomically-fixed but spatially-extended targets (abdominal organs), crops exclude large organ fractions. In both cases DSC collapses not because the model lacks global reasoning but because the crop removes the target itself. Window sensitivity within the model—where target visibility is unchanged—is the appropriate substitute. Additionally, the cross-task comparison is limited to two benchmarks (Synapse and Kvasir-SEG); the CS hypothesis should be validated on a broader range of medical segmentation tasks before being applied prescriptively. Finally, the three mechanisms proposed in Section V-C—inductive bias, over-context suppression, implicit regularization—are empirically motivated but not formally proven; their relative contributions and the conditions under which each dominates warrant further investigation.

VI. Conclusion

This paper presented ApproxDA, a controllable approximation framework derived from the dual-attention module of DA-TransUNet [2]. Rather than proposing another efficient attention operator, ApproxDA serves as a scientific instrument for investigating how approximation strength interacts with different medical image segmentation tasks.

Extensive experiments demonstrate that approximation effectiveness is fundamentally tied to task *Context Sensitivity (CS)*: the degree to which performance changes with the attention receptive field. With appropriate window selection, ApproxDA outperforms DA-TransUNet on both benchmarks: $+1.14\%$ DSC on Synapse multi-organ CT (80.94%, $M{=}28$) and $+1.73\%$ DSC on Kvasir-SEG polyp segmentation (90.17%, $M{=}56$). The $3.6\times$ sensitivity ratio (CS=2.30 pp vs. 0.64 pp) confirms that high-CS tasks require careful window tuning while low-CS tasks are window-robust. The non-monotonic Synapse DSC-vs-M curve—peak at intermediate $M{=}28$, declining at both extremes—mirrors the bias-variance tradeoff: under-context windows have high bias; over-context windows introduce spurious variance.

Furthermore, we identify three mechanisms explaining why approximation improves performance: inductive bias alignment (locality prior matching task structure), over-context suppression (reducing spurious distant correlations), and potential implicit regularization. Attention-map evidence directly supports the first: windowed PAM concentrates 62.3% of its effect on polyp regions vs. 34.2% for global attention ($1.82\times$ ratio).

Finally, our analysis reveals that learnable dual-branch routing collapses to $g{\approx}0.5$ regardless of window size or rank—a fundamental limitation of symmetric two-branch gating with broad implications for routing mechanism design.

Together, these findings shift the research question from *how to approximate* to *when, and at what scale*, with Context Sensitivity providing a practical task-level criterion. We hope this study motivates future work on CS quantification, broader validation across the segmentation task spectrum, and routing mechanisms that explicitly preserve branch diversity.

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